Observe that the graph of $y = \cos x$ looks just like the graph of $y = \sin x$ except it is being translated to the left by $\frac{\pi}{2}$. This is because of the identity $\cos x = \sin\left(\frac{\pi}{2} + x\right)$. It easily follows from the graph that $\cos x = \cos(-x) = \sin\left(\frac{\pi}{2} + x\right)$



(i) The sine and cosine functions are useful for one very important reason, since they repeat in a regular pattern (*i.e.*, they are periodic). There are a vast array of things in and around us that repeat periodically. For example, the rising and setting of the sun, the motion of a spring up and down, the tides of the ocean and so on, are repeating at regular intervals of time. All periodic behaviour can be studied through combinations of the sine and cosine functions.

(ii) Periodic functions are used throughout science to describe oscillations, waves and other phenomena that occur periodically.

Odd and Even trigonometric functions

Even and odd functions are functions satisfying certain symmetries. A real valued function $f(x)$ is an even function if it satisfies $f(-x) = f(x)$ for all real number x and an odd function if it satisfies $f(-x) = -f(x)$ for all real number x.
Basic trigonometric functions are examples of non-polynomial even and odd functions
Because $\cos(-x) = \cos x$ and $\sin(-x) = -\sin x$ for all x, it follows that $\cos x$ is an even function and $\sin x$ is an odd function.
Also note that $\sec x$ is an even function while $\tan x$, cosec x and $\cot x$ are all odd functions.
However, $f(t) = t - \cos t$ is neither even function nor odd function (why ?)

Example 3.14 Determine whether the following functions are even, odd or neither.

(i) $\sin^2 x - 2\cos^2 x - \cos x$ (ii) $\sin(\cos(x))$ (iii) $\cos(\sin(x))$ (iv) $\sin x + \cos x$

Solution:

(i) Let $f(x) = \sin^2 x - 2\cos^2 x - \cos x$
$f(-x) = f(x)$ [since $\sin(-x) = -\sin x$ and $\cos(-x) = \cos x$]
Thus, $f(x)$ is even.

(ii) Let $f(x) = \sin(\cos(x))$
$f(-x) = f(x)$, $f(x)$ is an even function.

(iii) $f(x) = \cos(\sin(x))$, $f(-x) = f(x)$, Thus, $f(x)$ is an even function.

(iv) Let $f(x) = \sin x + \cos x$
$f(-x) \neq f(x)$ and $f(-x) \neq -f(x)$
Thus, $f(x) = \sin x + \cos x$ is neither even nor odd.



(i) In general, a function is an even function if its graph is unchanged under reflection about the y-axis. A function is odd if its graph is symmetric about the origin.

(ii) The properties of even and odd functions are useful in analyzing trigonometric functions particularly in the sum and difference formula.

(iii) The properties of even and odd functions are useful in evaluating some definite integrals, which we will see in calculus.